

BABU BANARASI DAS UNIVERSITY



SCHOOL OF COMPUTER APPPLICATION

NO SQL PROJECT ON

Student Management System
Using MongoDB

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ACKNOWLEDGEMENT

I sincerely express my gratitude to my teacher **Mr. Harendra Singh Rajpoot** for giving me the opportunity to work on this NoSQL project titled “Student Management System Using MongoDB”.

This project helped me understand the basic concepts of NoSQL databases and learn how MongoDB is used to create, retrieve, update, sort and delete data. I also learned how comparison operators such as greater than, less than and equal to, along with logical operators such as AND and OR, are used in MongoDB.

I am thankful to my teacher for the guidance and support provided throughout the completion of this project.

INTRODUCTION

MongoDB is a popular NoSQL database management system that stores data in the form of documents. Unlike traditional relational databases, MongoDB does not require data to be stored in fixed rows and columns.

In this project, a Student Management System has been created using MongoDB. The database contains information about 50 students, including their roll number, name, age, marks, and city.

Different MongoDB operations have been performed on the student data, such as inserting, displaying, updating, deleting, sorting, counting, and searching records. Comparison operators such as \$eq, \$gt, \$lt, \$gte, and \$lte have also been demonstrated.

OBJECTIVES

To understand the concept of NoSQL databases.

- To learn the basic features of MongoDB.
- To create a database and collection.
- To insert 50 student records.
- To perform CRUD operations.
- To use comparison operators.
- To perform sorting and filtering.
- To understand AND and OR operations

1.CREATE & USE DATABASE

```
>_MONGOSH
```

```
> use("college")
```

```
< switched to db college
```

```
> db["students_"].find()
```

2. INSERT 50 STUDENTS DATA

```
>_MONGOSH
> db.Students2.insertMany([
  {
    "roll_no": 101,
    "name": "Aarav Mehta",
    "age": 19,
    "course": "BCA",
    "address": "Jaipur, Rajasthan"
  },
  {
    "roll_no": 102,
    "name": "Diya Kapoor",
    "age": 20,
    "course": "BSc CS",
    "address": "Pune, Maharashtra"
  },
  {
    "roll_no": 103,
    "name": "Kabir Malhotra",
    "age": 18,
    "course": "BBA",
    "address": "Delhi, Delhi"
  },
  {
    "roll_no": 104,
    "name": "Anaya Sharma",
    "age": 21,
    "course": "BCA",
    "address": "Indore, Madhya Pradesh"
  },
  {
    "roll_no": 105
```

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6ab40e65773f9c6a6284fab0'),
    '1': ObjectId('6ab40e65773f9c6a6284fab1'),
    '2': ObjectId('6ab40e65773f9c6a6284fab2'),
    '3': ObjectId('6ab40e65773f9c6a6284fab3'),
    '4': ObjectId('6ab40e65773f9c6a6284fab4'),
    '5': ObjectId('6ab40e65773f9c6a6284fab5'),
    '6': ObjectId('6ab40e65773f9c6a6284fab6'),
    '7': ObjectId('6ab40e65773f9c6a6284fab7'),
    '8': ObjectId('6ab40e65773f9c6a6284fab8'),
    '9': ObjectId('6ab40e65773f9c6a6284fab9'),
    '10': ObjectId('6ab40e65773f9c6a6284fab10'),
    '11': ObjectId('6ab40e65773f9c6a6284fab11'),
    '12': ObjectId('6ab40e65773f9c6a6284fab12'),
    '13': ObjectId('6ab40e65773f9c6a6284fab13'),
    '14': ObjectId('6ab40e65773f9c6a6284fab14'),
    '15': ObjectId('6ab40e65773f9c6a6284fab15'),
    '16': ObjectId('6ab40e65773f9c6a6284fab16'),
    '17': ObjectId('6ab40e65773f9c6a6284fab17'),
    '18': ObjectId('6ab40e65773f9c6a6284fab18')
```

3. SHOWING THE DATA

```
>_MONGOSH
> db.Students_.find()
< [
  '0': {
    name: 'Aarav Sharma',
    age: 20,
    course: 'BCA',
    address: {
      city: 'Lucknow',
      state: 'Uttar Pradesh'
    },
    marks: 82
  },
  '1': {
    name: 'Riya Verma',
    age: 19,
    course: 'BCA',
    address: {
      city: 'Kanpur',
      state: 'Uttar Pradesh'
    },
    marks: 76
  },
  '2': {
    name: 'Aditya Singh',
    age: 21,
    course: 'BTech',
    address: {
      city: 'Delhi',
      state: 'Delhi'
    },
    marks: 88
  }
]
```

4. Find Data By Condition (Find one)

```
db.Students2.findOne({ roll_no: 101 });  
  
{  
  _id: ObjectId('6ab54c4fad481e917eabbc5f'),  
  roll_no: 101,  
  name: 'Aarav Mehta',  
  age: 19,  
  course: 'BCA',  
  address: 'Jaipur, Rajasthan'  
}  
college>
```

5.Find Multiple Data By (Find Many)

```
>_MONGOSH
> db.Students2.find({
  age: { $eq: 20 }
})
< {
  _id: ObjectId('6ab54c4fad481e917eabb60'),
  roll_no: 102,
  name: 'Diya Kapoor',
  age: 20,
  course: 'BSc CS',
  address: 'Pune, Maharashtra'
}
{
  _id: ObjectId('6ab54c4fad481e917eabb65'),
  roll_no: 107,
  name: 'Vivaan Sethi',
  age: 20,
  course: 'BCA',
  address: 'Noida, Uttar Pradesh'
}
{
  _id: ObjectId('6ab54c4fad481e917eabb6a'),
  roll_no: 112,
  name: 'Sara Khan',
  age: 20,
  course: 'BCA',
  address: 'Lucknow, Uttar Pradesh'
}
{
  _id: ObjectId('6ab54c4fad481e917eabb6f'),
  roll_no: 117,
  course: 'Week 7 End of'
```

6.Greater Than(\$gt)

```
Welcome  >_ mongosh: Harsh  +
>_MONGOSH
> db.Students2.find({
  $and : [
    {age : { $gt : 20}},
    {course : "BTech"}
  ]
})
< {
  _id: ObjectId('6ab54c4fad481e917eabbc67'),
  roll_no: 109,
  name: 'Aditya Nair',
  age: 21,
  course: 'BTech',
  address: 'Kochi, Kerala'
}
{
  _id: ObjectId('6ab54c4fad481e917eabbc6c'),
  roll_no: 114,
  name: 'Meera Iyer',
  age: 21,
  course: 'BTech',
  address: 'Chennai, Tamil Nadu'
}
{
  _id: ObjectId('6ab54c4fad481e917eabbc71'),
  roll_no: 119,
  name: 'Dhruv Patel',
  age: 24,
  course: 'BTech',
  address: 'Surat, Gujarat'
}
```

7. Greater Than Equal (\$gte)

```
>_MONGOSH
> db.Students2.find({age : {$gte : 20}})
< {
  _id: ObjectId('6ab54c4fad481e917eabb60'),
  roll_no: 102,
  name: 'Diya Kapoor',
  age: 20,
  course: 'BSc CS',
  address: 'Pune, Maharashtra'
}
{
  _id: ObjectId('6ab54c4fad481e917eabb62'),
  roll_no: 104,
  name: 'Anaya Sharma',
  age: 21,
  course: 'BCA',
  address: 'Indore, Madhya Pradesh'
}
{
  _id: ObjectId('6ab54c4fad481e917eabb64'),
  roll_no: 106,
  name: 'Myra Joshi',
  age: 22,
  course: 'BSc CS',
  address: 'Dehradun, Uttarakhand'
}
{
  _id: ObjectId('6ab54c4fad481e917eabb65'),
  roll_no: 107,
  name: 'Vivaan Sethi',
  age: 20,
  course: 'BCA',
  address: 'Gurgaon, Haryana'
}
```

8. Less Than (\$lt)

> _MONGOSH

> db.Students2.find({age: {\$lt: 20}})

```
< [
  {
    _id: ObjectId('6ab54c4fad481e917eabbc5f'),
    roll_no: 101,
    name: 'Aarav Mehta',
    age: 19,
    course: 'BCA',
    address: 'Jaipur, Rajasthan'
  },
  {
    _id: ObjectId('6ab54c4fad481e917eabbc61'),
    roll_no: 103,
    name: 'Kabir Malhotra',
    age: 18,
    course: 'BBA',
    address: 'Delhi, Delhi'
  },
  {
    _id: ObjectId('6ab54c4fad481e917eabbc63'),
    roll_no: 105,
    name: 'Rohan Bansal',
    age: 19,
    course: 'BTech',
    address: 'Noida, Uttar Pradesh'
  }
]
```

>

9. Less Than Equal (\$lte)

```
>_MONGOSH
> db.Students2.find({age : {$lte : 20}})
< {
  _id: ObjectId('6ab54c4fad481e917eabbc5f'),
  roll_no: 101,
  name: 'Aarav Mehta',
  age: 19,
  course: 'BCA',
  address: 'Jaipur, Rajasthan'
}
{
  _id: ObjectId('6ab54c4fad481e917eabbc60'),
  roll_no: 102,
  name: 'Diya Kapoor',
  age: 20,
  course: 'BSc CS',
  address: 'Pune, Maharashtra'
}
{
  _id: ObjectId('6ab54c4fad481e917eabbc61'),
  roll_no: 103,
  name: 'Kabir Malhotra',
  age: 18,
  course: 'BBA',
  address: 'Delhi, Delhi'
}
{
  _id: ObjectId('6ab54c4fad481e917eabbc63'),
  roll_no: 105,
  name: 'Rohan Bansal',
  age: 19,
  course: 'BTech',
  address: 'Delhi, Delhi'
}
```

10. Not Equal (\$ne)

```
> _MONGOSH
> db.Students2.find({age: {$ne: 20}})
< [
  {
    _id: ObjectId('6ab54c4fad481e917eabbc5f'),
    roll_no: 101,
    name: 'Aarav Mehta',
    age: 19,
    course: 'BCA',
    address: 'Jaipur, Rajasthan'
  },
  {
    _id: ObjectId('6ab54c4fad481e917eabbc61'),
    roll_no: 103,
    name: 'Kabir Malhotra',
    age: 18,
    course: 'BBA',
    address: 'Delhi, Delhi'
  },
  {
    _id: ObjectId('6ab54c4fad481e917eabbc63'),
    roll_no: 105,
    name: 'Rohan Bansal',
    age: 19,
    course: 'BTech',
    address: 'Noida, Uttar Pradesh'
  }
]
```

11.IN (\$in)

>_MONGOSH

```
> db.Students2.find({  
  course : { $in : ["BCA", "BTech"]}  
})
```

```
< {  
  _id: ObjectId('6ab54c4fad481e917eabbc5f'),  
  roll_no: 101,  
  name: 'Aarav Mehta',  
  age: 19,  
  course: 'BCA',  
  address: 'Jaipur, Rajasthan'  
}  
{  
  _id: ObjectId('6ab54c4fad481e917eabbc62'),  
  roll_no: 104,  
  name: 'Anaya Sharma',  
  age: 21,  
  course: 'BCA',  
  address: 'Indore, Madhya Pradesh'  
}  
{  
  _id: ObjectId('6ab54c4fad481e917eabbc63'),  
  roll_no: 105,  
  name: 'Rohan Bansal',  
  age: 19,  
  course: 'BTech',  
  address: 'Chandigarh, Punjab'  
}  
{  
  _id: ObjectId('6ab54c4fad481e917eabbc65'),  
  roll_no: 107,  
  name: 'Vivaan Sethi',
```

12.AND OPERATOR (\$and)

```
Welcome  >_ mongosh: Harsh  +
>_MONGOSH
> db.Students2.find({
  $and : [
    {age : { $gt : 20}},
    {course : "BTech"}
  ]
})
< {
  _id: ObjectId('6ab54c4fad481e917eabbc67'),
  roll_no: 109,
  name: 'Aditya Nair',
  age: 21,
  course: 'BTech',
  address: 'Kochi, Kerala'
}
{
  _id: ObjectId('6ab54c4fad481e917eabbc6c'),
  roll_no: 114,
  name: 'Meera Iyer',
  age: 21,
  course: 'BTech',
  address: 'Chennai, Tamil Nadu'
}
{
  _id: ObjectId('6ab54c4fad481e917eabbc71'),
  roll_no: 119,
  name: 'Dhruv Patel',
  age: 24,
  course: 'BTech',
  address: 'Surat, Gujarat'
}
```

13.OR OPERATOR (\$or)

```
>_MONGOSH
> db.Students2.find({
  $or : [
    {age : {$gt : 22}},
    {course : "BTech"}
  ]
})
< {
  _id: ObjectId('6ab54c4fad481e917eabb63'),
  roll_no: 105,
  name: 'Rohan Bansal',
  age: 19,
  course: 'BTech',
  address: 'Chandigarh, Punjab'
}
{
  _id: ObjectId('6ab54c4fad481e917eabb67'),
  roll_no: 109,
  name: 'Aditya Nair',
  age: 21,
  course: 'BTech',
  address: 'Kochi, Kerala'
}
{
  _id: ObjectId('6ab54c4fad481e917eabb69'),
  roll_no: 111,
  name: 'Arjun Rao',
  age: 23,
  course: 'BSc CS',
  address: 'Hyderabad, Telangana'
}
{
```

14.NOR OPERATOR (\$nor)

```
>_MONGOSH
> db.Students2.find({
  $nor : [
    { age : { $gt : 20 } },
    { course : "BTech" }
  ]
})
< {
  _id: ObjectId('6ab54c4fad481e917eabb5f'),
  roll_no: 101,
  name: 'Aarav Mehta',
  age: 19,
  course: 'BCA',
  address: 'Jaipur, Rajasthan'
}
{
  _id: ObjectId('6ab54c4fad481e917eabb60'),
  roll_no: 102,
  name: 'Diya Kapoor',
  age: 20,
  course: 'BSc CS',
  address: 'Pune, Maharashtra'
}
{
  _id: ObjectId('6ab54c4fad481e917eabb61'),
  roll_no: 103,
  name: 'Kabir Malhotra',
  age: 18,
  course: 'BBA',
  address: 'Delhi, Delhi'
}
{
```

15.NOT OPERATOR (\$not)

```
>_MONGOSH
> db.Students2.find({
  age : {$not : {$gt : 20}}
})
< {
  _id: ObjectId('6ab54c4fad481e917eabb5f'),
  roll_no: 101,
  name: 'Aarav Mehta',
  age: 19,
  course: 'BCA',
  address: 'Jaipur, Rajasthan'
}
{
  _id: ObjectId('6ab54c4fad481e917eabb60'),
  roll_no: 102,
  name: 'Diya Kapoor',
  age: 20,
  course: 'BSc CS',
  address: 'Pune, Maharashtra'
}
{
  _id: ObjectId('6ab54c4fad481e917eabb61'),
  roll_no: 103,
  name: 'Kabir Malhotra',
  age: 18,
  course: 'BBA',
  address: 'Delhi, Delhi'
}
{
  _id: ObjectId('6ab54c4fad481e917eabb63'),
  roll_no: 105,
  name: 'Rohan Bansal',
```

16.CURSOR OPERATOR (limit)

>_MONGOSH

> db.Students2.find().limit(5)

```
< {
  _id: ObjectId('6ab54c4fad481e917eabb5f'),
  roll_no: 101,
  name: 'Aarav Mehta',
  age: 19,
  course: 'BCA',
  address: 'Jaipur, Rajasthan'
}
{
  _id: ObjectId('6ab54c4fad481e917eabb60'),
  roll_no: 102,
  name: 'Diya Kapoor',
  age: 20,
  course: 'BSc CS',
  address: 'Pune, Maharashtra'
}
{
  _id: ObjectId('6ab54c4fad481e917eabb61'),
  roll_no: 103,
  name: 'Kabir Malhotra',
  age: 18,
  course: 'BBA',
  address: 'Delhi, Delhi'
}
{
  _id: ObjectId('6ab54c4fad481e917eabb62'),
  roll_no: 104,
  name: 'Anaya Sharma',
  age: 21,
  course: 'BCA',
```

17.CURSOR OPERATOR (sort)

```
>_MONGOSH
> db.Students2.find().sort({age : 1})
< {
  _id: ObjectId('6ab54c4fad481e917eabbc6b'),
  roll_no: 113,
  name: 'Reyansh Gupta',
  age: 18,
  course: 'BBA',
  address: 'Kanpur, Uttar Pradesh'
}
{
  _id: ObjectId('6ab54c4fad481e917eabbc66'),
  roll_no: 108,
  name: 'Ishita Arora',
  age: 18,
  course: 'BBA',
  address: 'Amritsar, Punjab'
}
{
  _id: ObjectId('6ab54c4fad481e917eabbc61'),
  roll_no: 103,
  name: 'Kabir Malhotra',
  age: 18,
  course: 'BBA',
  address: 'Delhi, Delhi'
}
{
  _id: ObjectId('6ab54c4fad481e917eabbc7c'),
  roll_no: 130,
  name: 'Ira Mukherjee',
  age: 18,
  course: 'BSc CS',
```

18. CURSOR OPERATOR (skip)

```
>_MONGOSH
> db.Students2.find().skip(5)
< {
  _id: ObjectId('6ab54c4fad481e917eabbc64'),
  roll_no: 106,
  name: 'Myra Joshi',
  age: 22,
  course: 'BSc CS',
  address: 'Dehradun, Uttarakhand'
}
{
  _id: ObjectId('6ab54c4fad481e917eabbc65'),
  roll_no: 107,
  name: 'Vivaan Sethi',
  age: 20,
  course: 'BCA',
  address: 'Noida, Uttar Pradesh'
}
{
  _id: ObjectId('6ab54c4fad481e917eabbc66'),
  roll_no: 108,
  name: 'Ishita Arora',
  age: 18,
  course: 'BBA',
  address: 'Amritsar, Punjab'
}
{
  _id: ObjectId('6ab54c4fad481e917eabbc67'),
  roll_no: 109,
  name: 'Aditya Nair',
  age: 21,
  course: 'BTech',
```

19. MONGODB OPERATOR USED

Operator	Meaning	Example
\$eq	Equal to	marks: {\$eq:88}
\$gt	Greater than	marks: {\$gt:85}
\$lt	Less than	marks: {\$lt:60}
\$gte	Greater than or equal to	marks: {\$gte:90}
\$lte	Less than or equal to	marks: {\$lte:70}
\$and	All conditions true	\$and:[...]
\$or	At least one condition true	\$or:[...]

20.CONCLUSION

This project provided practical knowledge of NoSQL databases using MongoDB. A Student Management System containing student information was created and different MongoDB operations were performed on the data.

Through this project, I learned how to insert, retrieve, update and delete documents. I also learned how ascending and descending sorting works, how comparison operators such as \$eq, \$gt, \$lt, \$gte and \$lte work, and how logical operators such as \$and and \$or can be used to filter data.

Overall, this project helped me understand how MongoDB can be used to manage and query student data efficiently.

21. REFERENCES

- a. MongoDB documentation and MongoDB shell concepts.
- b. Babu Banarasi Das University NoSQL practical/reference document supplied for this project.
- c. Course notes and classroom guidance.

